

Name	Subject Class	Class	Candidate Number
	2BI		



ANGLO-CHINESE JUNIOR COLLEGE
H2 Biology Preliminary Examinations 2017

H2 BIOLOGY
 Practical

9744/04
 02 Aug 2017
2 hours and 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, index number, class, shift and laboratory on this Question Paper.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	/21
2	/20
3	/14
Total	55

This question paper consists of 18 printed pages.

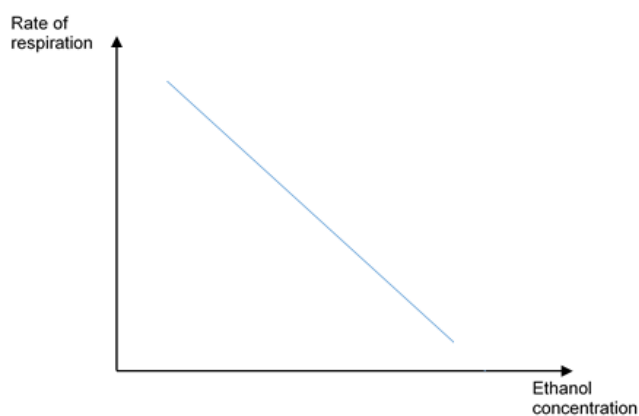
- 1 During anaerobic respiration, yeast cells use glucose and release carbon dioxide and ethanol. Ethanol is known to disrupt enzymes involved in the respiration pathway.

In this experiment, you will investigate the effect of different concentrations of ethanol on the rate of respiration in yeast cells.

- (a) Sketch a fully-labelled graph to show the expected relationship between the rate of respiration and the concentration of ethanol, as ethanol concentration increases.

Explain the shape of your graph.

[4]



1. Y-axis labelled rate of respiration and x-axis labelled ethanol concentration;
2. Line / Curve with negative gradient;

3. As ethanol concentration increases, rate of respiration decreases;

4. Active site of enzyme is lost/disrupted with one possible reason, e.g. competitive/non-competitive inhibitor, 3D conformation of enzyme is changed;

5. Enzymes in glycolysis/respiration cannot form enzyme-substrate complexes;

Max. 2

Methylene blue is a dye which, under certain conditions, is easily reduced to a colourless compound. In the presence of reduced NAD which is produced during glycolysis, methylene turns from blue to colourless.

In your investigation, you are to dilute the given ethanol solution to obtain different concentrations of ethanol solution.

You are provided with:

- 30 cm³ of 5.0% yeast solution, **Y**
- 20 cm³ of 100% ethanol, **E**
- 25 cm³ of glucose, **G**
- 6 cm³ of methylene blue, **M**

Proceed as follows:

1. Label 5 boiling tubes 1, 2, 3, 4, 5. Place 5 cm³ of suspension **Y** into each of the boiling tubes. Ensure **Y** is well-suspended. Place all the boiling tubes into a water bath of 60°C for 5 minutes.
2. While waiting, carry out a dilution to make up 5 cm³ of different concentrations of ethanol solutions using the vials provided. When preparing the solutions, add distilled water first before adding ethanol. This is to ensure better mixing of ethanol and distilled water.

Complete Table 1.1 to show how you will make the different concentrations of ethanol solution.
[2]

Table 1.1

vial	concentration of ethanol / %	volume of distilled water / cm ³	volume of ethanol / cm ³
1	100	0.0	5.0
2	80	1.0	4.0
3	60	2.0	3.0
4	40	3.0	2.0
5	20	4.0	1.0

1. Correct concentration (Equal intervals across an appropriate range – max lowest concentration = 40%, R! 0%);
2. Correct volumes and precision (1dp);

R! total volume less or more than 5.0 cm³

Read through steps 3 to 10 to prepare a table to record your results in (b), before starting the investigation.

3. After 5 minutes, lower the temperature of the water bath to 45°C. Leave the boiling tubes in the water bath for 1 minute.
4. Place 2 cm³ of the ethanol solution from vial 1 into boiling tube 1 then add in 2 cm³ of G.
5. Use a dropper to add 1 cm³ of M into boiling tube 1 immediately.
6. Shake the boiling tube sufficiently to mix the contents well. The mixture should turn pale blue.
7. Carefully place the boiling tube back into the water bath and start the stopwatch. Do not shake or stir the boiling tubes from this point onwards as it may affect the results.
8. Stop the stopwatch when the blue mixture has been decolourised. The surface of the mixture in the boiling tube may remain blue. Record the time taken and hence the rate of respiration in the table prepared in (b). If the mixture does not decolourise within 12 minutes, record 'more than 720' as the time taken and the rate as '0'.
9. Repeat steps 4 to 8 with boiling tubes 2, 3, 4, and 5, in turn.
10. After completing the experiment, shake boiling tube 5 vigorously about 10 times. Record your observations in part (d).

(b) Record your results for each concentration of ethanol in a suitable format in the space below.

[4]

Table showing effects of ethanol concentration (%) on the rate of respiration of yeast (%)

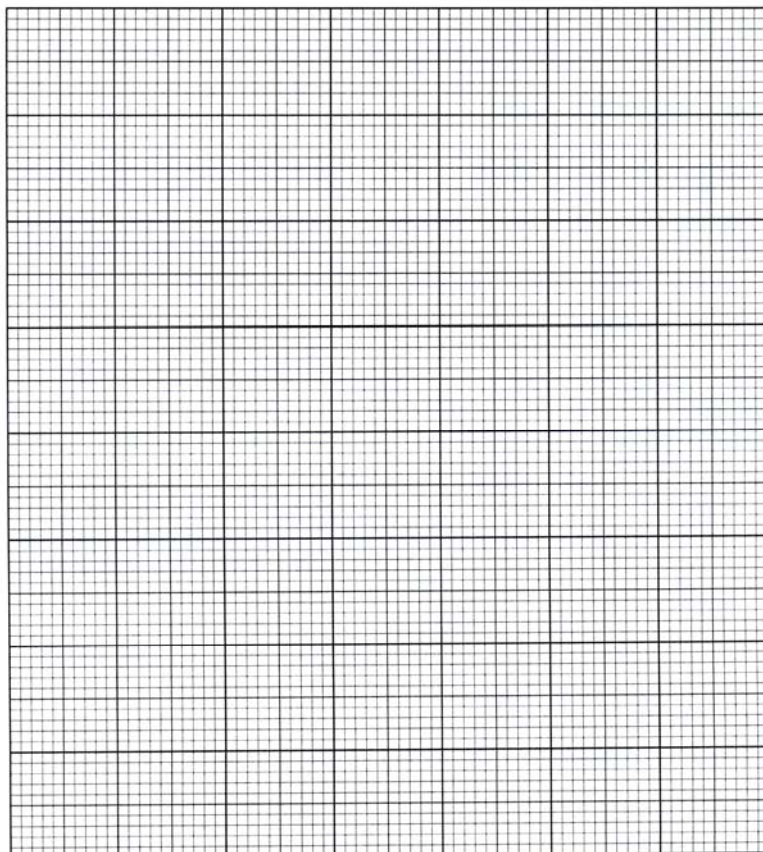
Boiling tube	Ethanol concentration / %	Time taken / s	Rate of respiration / s ⁻¹
1	100		
2	80		
3	60		
4	40		
5	20		

1. Table layout: Independent variable to occupy leftmost column;
2. Appropriate column headings + Units;
3. Precision of data: Raw data: to the nearest second, rate of respiration: 3sf if time taken is in hundreds (consistency);
4. Complete set of data (including rate – increasing rate as ethanol concentration decrease) + Correct calculation + correct trend (accept more than 720 for 100% and 80%);

For tubes with 'more than 720', since rate is $1/\infty$, hence rate is 0.
Reject '-'

(i) Use the grid below to display your results from (b).

[4]



1. Independent variable (concentration of ethanol) on x-axis (no need to mark for units);
2. Both axes correctly labelled with units (concentration of ethanol/% on the rate of respiration/s⁻¹);
3. Sensible scale with graph occupying at least ½ the grid on both x- and y- axes + equidistant divisions on axes + accurate plotting of points to nearest half a small square;
4. Line of best fit drawn, without extrapolation beyond plotted points (appropriate where data gives confidence in underlying relationship) / dot-to-dot plot joined by a straight line;

(ii) Discuss what these results suggest about the relationship predicted in part (a). [2]

1. No clear pattern in the results/results do not match at high ethanol concentrations
2. Decreases confidence in the predicted relationship / further results will need to be collected in order to further evaluate the relationship

OR (if experiment was conducted and matches predicted results)

1. Pattern of results shows same pattern as predicted in (a);
2. Which increases the confidence in the hypothesis / proposed relationship;

OR (if pattern of results initially matches predicted pattern if plateau is present)

1. Pattern of results shows same pattern as predicted in (a) initially, but no plateau is reached;
2. Decreases confidence in the predicted relationship / further results will need to be collected in order to further evaluate the relationship;

R! confirming results... / concluded that results are... / results are true/valid

(c) State and explain your observations from step 10. [1]

1. Mixture turns blue again as oxygen was reintroduced into the mixture, reoxidising methylene blue;

(d) Suggest how adding glucose solution to the mixture increases the validity of the results. [2]

1. Glucose is required (as the raw material / substrate) for respiration / glycolysis;
2. So substrate / glucose is not / less likely to be limiting;

3. The concentration of ethanol would thus be the only variable in the experiment / Changes in rate of respiration will be due to changes in concentration of ethanol;

(e) One way to increase confidence in the conclusions of this investigation would be to repeat the experiment several times.

Describe two other modifications to the method that would increase confidence in the conclusions, and explain how these modifications would achieve this. [2]

1. Repeat with a control using boiled and cooled yeast to check if the results are due to (anaerobic) respiration of yeast;
2. Repeat with a control using 0% of ethanol solution / 2 cm³ of distilled water to check if the results are due to presence of ethanol;
3. Add methylene blue before adding glucose to prevent problem of yeast using glucose for respiration before time taken for decolourisation is recorded;
4. Use a thermostatically-controlled water bath to ensure the temperature is kept constant;
5. Use a colourimeter to determine end-point (to measure absorbance of solutions at regular intervals);

6. Use a lid/parafilm to minimize changes in concentration of ethanol solutions due to evaporation of water;
(Note: rate of evaporation is affected to a similar extent for most experiments involving solutions; however, for this experiment, the range of ethanol concentration is large and discrepancies in evaporation might be magnified)
-
7. Repeat experiment with smaller intervals of ethanol concentration, e.g. 10% difference between each solution;
-

[Total: 21]

2 Section A

You are required to carry out an investigation to estimate the water potential (ψ) of the cells of the plant material with which you have been provided.

You are provided with stems of a plant sample and different concentrations of sucrose solution.

Proceed as follows:

1. Using a sharp scalpel, cut a 5 cm long, straight piece, from near the middle region, of one of the specimens provided. Hold this piece of plant in a vertical position and cut it longitudinally downwards for a distance of approximately 4 cm (Fig. 2.1).

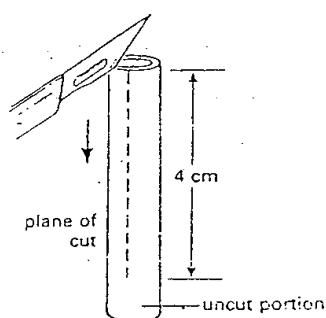


Fig. 2.1

2. You should find that the specimen has curved as shown in Fig. 2.2. Check that the distance A, between the cut pieces is at least 1 cm. If not, repeat the procedure using another specimen. Place the piece of plant tissue horizontally in the base of a clean and dry petri dish. Taking care not to squash the plant material, gently but firmly fix it to the dish using a small roll of plasticine, which you press down at X and Y (Fig. 2.2).

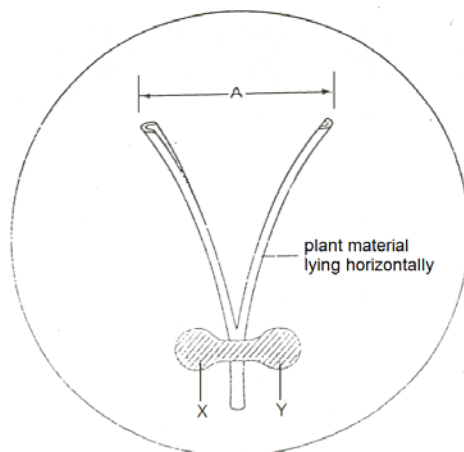


Fig. 2.2

3. Prepare **three** further dishes, using 5 cm long pieces of tissue, cut from roughly corresponding positions of three other stalks. Label your dishes 1, 2, 3 and 4.

4. Place the four dishes on the separate sheet of graph paper provided and measure, to the nearest millimeter, the distance A in each dish. Record these observations in the table below. [2]

	Dish 1 (for S1)	Dish 2 (for S2)	Dish 3 (for S3)	Dish 4 (for S4)
Initial value of A / mm				
Value of A after 10 minutes / mm				
Difference between initial and final value of A / mm				
Change / %				

1. Correct precision for A: whole number + Correct precision for percentage change: 1/2 sf;
2. Complete set of data + Correct trend: more negative percentage change (negative gradient);

5. You have been provided with the following sucrose solutions:

S1 is 0.2 mol dm⁻³

S2 is 0.4 mol dm⁻³

S3 is 0.6 mol dm⁻³

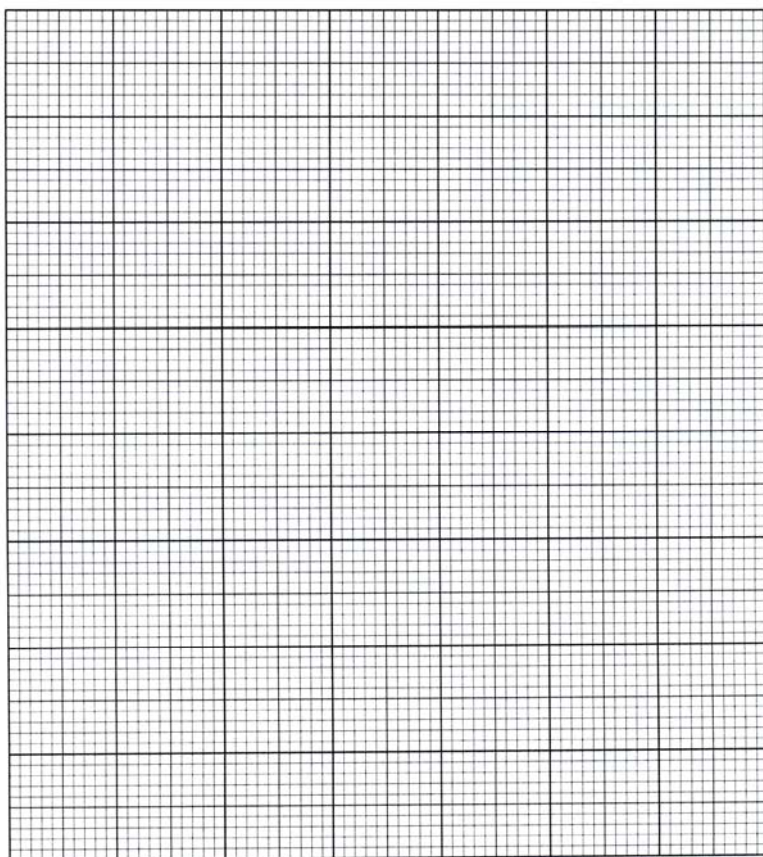
S4 is 0.8 mol dm⁻³

6. Gently, in order to avoid dislodging the plant tissue, pour **S1** into **dish 1**, so that the piece of plant is completely covered by the solution. As quickly as possible, pour the other solutions into their respective dishes.
7. Leave the dishes for 10 minutes. During this time you may begin with Section B.
8. After 10 minutes, measure (to the nearest mm) the distance A in each of the dishes. Record these measurements in the table in Step 4.
9. For each dish, calculate the percentage change in A, and also record this in the table (Step 4). State, in each case, if the value is positive or negative.

10. Plot a graph of the percentage change in A against molarity of sucrose solution.

[3]

1. Independent variable (sucrose concentration) on x-axis (no need to mark for units) + Both axes correctly labelled with units (percentage change in A (%) against sucrose concentration (mol dm^{-3});
2. Sensible scale with graph occupying at least $\frac{1}{2}$ the grid on both x- and y- axes + equidistant divisions on axes;
3. Line of best fit drawn, without extrapolation beyond plotted points (appropriate where data gives confidence in underlying relationship) / dot-to-dot plot;



11. The following table shows the solute potentials (ψ_s) of different concentrations of sucrose solutions, at the approximate temperature at which you have been working.

Concentration / mol dm ⁻³	Solute potential (ψ_s) / kilopascals
0.1	-260
0.2	-540
0.3	-820
0.4	-1120
0.5	-1450
0.6	-1700
0.7	-2170
0.8	-2580

12. Use the graph you have drawn, and the table above, to estimate the solute potential of the cells of this plant material. Explain fully how you arrived at your answer. [3]

Answer: **1. Accept solute potential from -260 to -820 ψ_s ;
R! water potential**

Explanation: **2. There is no change when [sucrose] = ____ mol dm⁻³; (refer to Q10 graph);**

3. It is the point at which there is no net movement of H₂O via osmosis;

(4. Given that $\psi_w = \psi_s + \psi_p$, the cells here are at incipient plasmolysis, where ψ_p of the cell = 0 kPa;)

(Point 4 is not in the syllabus) @ 1m

Section B

In this section, you will require access to a microscope and slide **K1**.

K1 is a stained, longitudinal section of a young onion root tip in which some cells are undergoing mitosis. Fig. 2.3 shows a plan diagram of **K1**.

Examine **K1** carefully, in the region labelled **A** in Fig. 2.3, using low- and high- power objectives of your microscope.

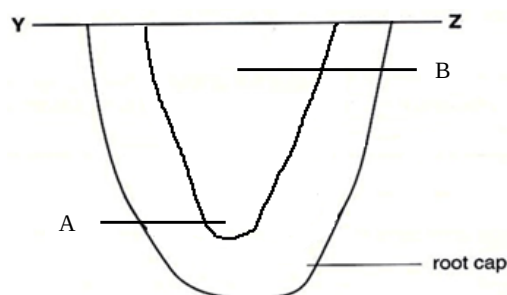


Fig. 2.3

1. Make a labelled, high-power drawing of a cell in anaphase from region **A**. [4]

Accuracy of drawing;	1. Show all the structures (chromosomes, plasma membrane, cytoplasm, cell wall) that can be seen in the defined part of a specimen; R! double-arm structure of chromosomes 2. Direction of chromosome separation: to opposite poles;
Clarity of drawing;	1. Use of sharp HB pencil 2. Clear single neat lines
Scale of drawing;	1. Use at least 2/3 of space provided 2. Correct proportion (Chromosomes and cell is drawn to the same scale - same magnification)
Label of drawing;	1. Clear straight lines 2. Correct labels (chromosomes*, plasma membrane, cytoplasm, cell wall) * Compulsory point

2. Using the eyepiece graticule fitted in the eyepiece lens of your microscope, and the stage micrometer, find the actual length, in μm , of the cell that you have drawn.

Show the measurements that you made and your working. [3]

1. **PDO: Shows division of stage micrometer measurement by number of eyepiece graticular divisions (+ what power is used);**
2. **MMO: Shows measurement of cells from slide in eyepiece graticular division (5 to 23);**
3. **ACE: Conversion of measurement from graticular division to answer in μm ;**

Length of cell = _____ μm

3. Measure and calculate the average length of the cells from both regions A and B. Record your results and measurements in a suitable table below. [2]

1. **At least 3 counts for each region in whole numbers + Average in whole number/1 dp;**
2. **Length of A is between 12.5-57.5 μm , length of B is between 37.5-75 μm ;**

4. Decide a statistical test that you can use to determine if there is a significant difference between the length of the cells in regions A and B. [1]

T test;

 _____ @ 1m

5. A student made some measurements of the length of cells of a garlic root tip undergoing mitosis in regions A and B. A summary of the student's results is shown in Table 2.1.

Table 2.1

Length of Cells / μm		Significance of difference
Region A	Region B	
32	58	$P < 0.05$

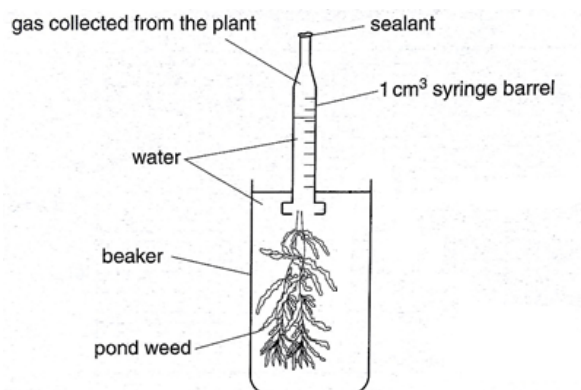
Comment on what these results show and suggest an explanation for any pattern. [2]

1. * Results shows that there is a significant difference between the length of cell B and A where B is longer in length than A;
2. Cells in region A are the resultant daughter cell after mitosis / Cells in region B have not yet undergone mitosis;
3. Mitosis in plant cell involved the formation of a cell plate via vesicles from the Golgi apparatus (instead of cell elongation followed by cleavage furrow in cytokinesis of animal cells);
4. Hence cells in region A is about half the length of cells in region B;

* Compulsory point

[Total: 20]

- 3 The rate of photosynthesis can either be measured by the rate at which carbon dioxide is taken in or the amount of oxygen that is given out. Some water plants release bubbles of gas from a freshly cut stem when illuminated. Light intensity is controlled using five filters, F1, F2, F3, F4, F5.



Different water plants are adapted to different light intensities. A sun-loving water plant is adapted to high light intensities while a shade-loving water plant is adapted to low light intensities.

Using this information, the set-up above and your own knowledge, design an experiment to investigate the effect of light intensity on photosynthesis in sun and shade plants.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you **must** use:

- Sun plant
- Shade plant
- Bench lamp with 60 W bulb
- 5 filters (F1, F2, F3, F4, F5) which can be adjusted to allow different amounts of light to pass through
- 1% sodium hydrogencarbonate solution

You may select from the following apparatus and use appropriate additional apparatus:

- Normal laboratory glassware, e.g. a variety of different sized beakers, measuring cylinders, and syringes for measuring volumes
- Forceps
- Timer, e.g. stopwatch

Your plan should:

- have a clear and helpful structure such that the method you use is repeatable by anyone reading it
- be illustrated by relevant diagram(s), if necessary, to show, for example, the arrangement of the apparatus used
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimize any risks associated with the proposed experiment.

[Total: 14]

Marking scheme

Compulsory

1	Broad outline C 1m	General idea <i>How would you measure the dependent variable?</i> (Oxygen is given off during light dependent stage via photolysis.) The experiment involves measuring oxygen evolved in response to different light intensities over a set period of time. OR Mark will be awarded if the broad outline of experiment is reflected in the procedure.																																																																									
2	Independent variable C 1m	State what the independent variable is, use at least five different values with regular intervals, good range, units <u>Independent</u> variable: Light intensity (20%, 40%, 60%, 80%, 100% transmission) A! Highest low limit: 30% Lowest high limit: 90% A! Lux R! Au																																																																									
3	Method (How to Vary IV) C 1m	Plan suitable method to vary the independent variable. Place filters with the different light transmissions in front of the lamp.																																																																									
4	Table C 1m	Shows how results are to be presented in the form of a table with independent and dependent variables in appropriate columns / rows. Units must be correct. <table><tr><th rowspan="2">Type of plant</th><th rowspan="2"><u>Light intensity</u> / %</th><th colspan="4">Distance moved / Volume of oxygen evolved in 10 min/cm³</th><th rowspan="2"><u>Average</u> rate of photosynthesis/ Average rate of oxygen evolved / cm³ s⁻¹</th></tr><tr><th>Try 1</th><th>Try 2</th><th>Try 3</th><th>Average</th></tr><tr><td rowspan="5">Sun plant</td><td>20</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>40</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>60</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>80</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>100</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td rowspan="5">Shade plant</td><td>20</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>40</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>60</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>80</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>100</td><td></td><td></td><td></td><td></td><td></td></tr></table> A! Separate table for sun and shade plant No need control Units: To follow raw data collected	Type of plant	<u>Light intensity</u> / %	Distance moved / Volume of oxygen evolved in 10 min/cm ³				<u>Average</u> rate of photosynthesis/ Average rate of oxygen evolved / cm ³ s ⁻¹	Try 1	Try 2	Try 3	Average	Sun plant	20						40						60						80						100						Shade plant	20						40						60						80						100					
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Commented [THA1]: To teach students considerations when deciding range

5	Risk/ safety C 1m	Risk / safety What are the hazard and precaution? 1. Take care when cutting plants. Use forceps to hold plants so as to prevent cutting fingers; 2. Sodium hydrogen carbonate is an irritant. Wear gloves when handling; 3. Avoid touching the electrical socket with wet hands as there may be a risk of electrocution; 4. Bulb of lamp will be hot, do not touch when lamp is in use; 5. (<i>Low priority</i>) Take care when handling glassware to prevent breakage which may cause injury;
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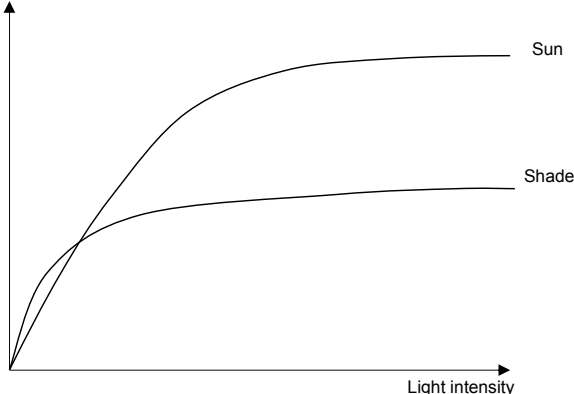
Max 9m from any of the below points

6	Method + Scientific reasoning (Overall) 1m	Plan a suitable method that involves monitoring/measuring the DV in response to the varied ID over a period of time/ set interval <i>If experiment is fundamentally wrong, do not award for this mark.</i> Theory (1m) <i>How would the independent variable affect the dependent variable?</i> An increase light intensity will increase the rate of photophosphorylation as more electrons are excited. In order to fill the electron gap, the rate of photolysis of water increases, where water is broken down to form of H ⁺ and oxygen, which is given off . The higher the light intensity, the higher the rate of oxygen formation. (The sun plant would have higher rate of photosynthesis at high light intensities while shade plants would have a higher rate of photosynthesis at low light intensities.)
7	Dependent variable (1m)	State what is the dependent variable. Dependent variable: (Rate of photosynthesis measured by) volume of oxygen evolved / distance of meniscus moved;
8	Method (1m) How to measure / monitor DV	Specifies method of measuring / monitoring DV Record the change in liquid level / volume of gas over a period of (10 min);
9	Controlled variables to improve accuracy or reliability (1m)	Identifies at least two variables to control; (Mark if specified in diagram) Controlled variable 1. Distance of lamp from plant; 2. Distance of filter from lamp; 3. Length of plant / number of leaves / number of plants / size of leaves; 4. Duration of exposure to light; 5. Volume of sodium hydrogen carbonate added to the water; 6. Temperature; R! Concentration of sodium hydrogen carbonate (Given in question)

Commented [THA2]: To teach in class: Write controlled variable

Commented [THA3]: Accept "amount" but circle and highlight

10/11	Controlled variables (2m)	Describes how two identified variables are controlled. (Must specify appropriate value) 1. Place a lamp of 60 W bulb at a distance of 15 cm away from the beaker of water; 2. Cut a branch of plant 2 cm long and place it in a glass filter funnel; 3. Turn on the lamp for a period of 10 min; 4. Add 2 cm³ of sodium hydrogencarbonate added; 5. Use a thermostatically-controlled water bath;
12	Control (1m)	Conduct a control or * experiment using boiled and cooled sun and shade plant / no plant / plant without leaves (*with the same set up / experimental conditions.) (This is to ensure that the changes in volume of oxygen is due to the change of rate of photosynthesis.) R! Zero light intensity is not a good control because gas will still be produced. (Specific to photosynthesis)
13/14	Method (2m)	Steps to take that would ensure the validity of the experimental results. (Not the same as controlled variable) 1. Adding excess sodium hydrogen carbonate into the water (to ensure sufficient concentration of CO₂). 2. Use a fresh solution of sodium hydrogencarbonate for each replicate (to ensure sufficient concentration of CO₂); 3. Conduct experiment in a dark room / eliminate all other light sources / ensure all other light sources are constant (to prevent heating effect); 4. Use a cool light source / water screen in front of the lamp / use thermostatically-controlled water bath (to prevent heating effect of lamp)*<i>unless about thermostatically controlled water bath</i>; 5. Ensure the water level of thermostatically-controlled water bath is above the water level in the beaker (to ensure homogenous temperature of liquid); 6. Pick actively bubbling plants (for observable displacement of water) 7. Ensure sufficient number of leaves (for observable displacement of water); 8. AVP;
15	Method (1m)	Plan a method for equilibration Immerse the plant in the sodium hydrogen carbonate solution and illuminate it for fixed time (10) minutes for equilibration before starting to collect oxygen;
16	Reliability (1m)	Reference to repeating at least two more time with different experimental subjects. Perform experiment for another 2 times using another branch / plant;

17	Accuracy (1m)	Repeating experiment with smaller intervals Repeating the experiment with filters of smaller interval of light transmission at 10% intervals / any suitable interval;
18	Graph (1m)	Axes drawn must be correct, two graphs on the same axes Average rate of photosynthesis  Look out for • Shape: Shade > Sun at low light intensities, Sun > Shade at high light intensities, Plateau on both graphs • A! Rate of photosynthesis and light intensity